Fiscal Year:	FY 2010	**Task Last Updated:**	FY 07/01/2010
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Project Title:	Patterns of Energy Deposition by HZE Particles in Cellular Targets		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	07/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: New end date is 7/31/2011 per C. Guidry/JSC (6/2010) NOTE: New end date is 7/31/2010, per PI (9/2009) NOTE: Received NCE to 7/31/2009 per PI (1/2009) NOTE: End date changed to 12/31/2008 per S. Krenek/JSC (8/07) End date changed to 7/31/2008 per JSC info update (10/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Toburen, Larry (East Carolina University)		
Grant/Contract No.:	NNJ04HF39G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The interpretation of data from mechanistic studies on radiation biology requires detailed descriptions of the local patterns of energy deposition in the appropriate cellular or sub-cellular sites to explore the effects of radiation quality on biological response. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The calculations focus on interpretation of microscopic “event” or “dose” distributions appropriate to recently initiated studies of HZE radiation induced leukemia in animal and cellular models [NSCOR Program, Colorado State University, R. Ulrich, PI] and those encountered in the production of clustered damage in DNA [e.g., Brookhaven National Laboratory, B Sutherland, PI]. The progenitor cells for leukemia are haematopoietic stem cells that are located in clusters within marrow cavities having dimensions on the order of a few hundred microns. These cavities are surrounded by trabecular bone matrix, which forms a wall of several hundred microns thickness around the marrow. The dimensions and geometry of the bone marrow are too irregular for direct dosimetry measurements and dose-averaging methods are inappropriate for chronic low dose-rate (single particle) exposures. Detailed Monte Carlo charged particle track structure simulation will provide quantitative information on spatial patterns of energy deposition in cellular targets of dimensions as small as a few nanometers on a particle-by-particle basis; these distributions are the starting point to characterize the detailed nature of radiation-induced damage produced by high-LET radiation. The modeling approach will combine the capabilities of full event-by-event Monte Carlo track simulation codes with codes employing condensed history capabilities; the latter is more appropriate for the description of high-energy secondary electrons. The event-by-event ion code PROTRAC (which is part of the biophysical simulation tool PARTRAC) developed at GSF Neuherberg for protons and alpha particles will be extended for heavy charged particle (HZE) interactions using appropriate relativistic theory. Generated secondary electrons that traverse specific regions of interest, e.g., bone marrow cells or their nuclei, will be followed also event-by-event using ETRAC (the electron transport code of PARTRAC) whereas electron transport in adjoining tissues and the surrounding structural material will be followed by the more computer efficient condensed history general purpose MC transport code PENELOPE developed at the University of Barcelona. Appropriate interaction cross sections for a wide range of target materials will be added to the ETRAC and PROTRAC codes, specifically to explore the effects of bone-tissue interfaces on energy deposition in bone marrow cells. Comparison of the spatial pattern of energy deposited in volumes of interest obtained from models of the HZE particles with that obtained from low-LET radiation will provide a firm basis to interpret radiation quality effects on biological response and the mathematical structure developed is amenable to scaling to volumes appropriate to human dimensions for considerations of risk assessment. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The focus will be on Leukemia, i.e., on the bone/bone-marrow interface from realistic mouse or human dimensions down to single cells. Little is known on the interaction of HZE particles with the human body. HZE particles are part of the Galactic Cosmic Rays (GCR) which exist in deep space and cannot be shielded completely inside spacecrafts. The latency time of leukemia, 2-3 years, coincides more or less with the mission time for the planned expedition to Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Progress has been made in obtaining interaction cross sections for protons and HZE particles with biologically relevant substances. We are currently updating inelastic proton interaction cross sections with liquid water which serves as substitute of soft tissue and bone marrow. We have started the derivation of proton interaction cross sections with metallic calcium, a major component of bone, which serves as substitute for bone in PARTRAC. We have adopted a model for the optical dielectric response function of calcium based on theoretical models and available experimental information. The model fulfills sum rules and reproduces other properties of the optical and dielectric functions. The calculated mean excitation energy is I = 172 eV, which is around 10 % smaller than the ICRU recommended value of 191 ± 8 eV. We are currently deriving the energy loss function with the full dependence on energy and momentum transfer. We are exploring different extension algorithms with the aim of balancing accuracy with applicability. Our current model includes a separation of the energy loss functions into subshells. Parameters are determined using theoretical constraints of the model to eliminate uncertainties. In order to calculate interaction cross sections and stopping powers we need to model the Bethe surface of calcium and the separation of sub-shells in an effective way. We are using a Gaussian model to represent different electron groups (shells) and use a parameter-free model to determine key parameter in modeling the Bethe surface. Models for the extension algorithm are in agreement with published and widely used models. Since we have problems in calculating interaction cross sections at relativistic energies – calculations agree well with expectations for non-relativistic energies; however they deviate significantly at relativistic energies using a plane-wave Born approximation. To get a better insight into this problem we repeated our procedure of modeling the Bethe surface for silicon. The optical oscillator strength as well as calculations of the mean free path and stopping powers within the simpler Bethe theory are published (Fernández-Varea et al, Nucl. Instr. Meth. B 229 (2005) 187-218). Furthermore, some experimental data exist for electron impact on silicon at intermediate energies. Therefore, we have taken the optical oscillator strength from the literature and modeled the Bethe surface using our Gaussian approach with a standard extension algorithm. We have calculated electron interaction cross sections using different plane-wave approaches and different integration techniques. We get good agreement with other calculations for the mean free path and stopping power at moderate energies, but under- (over-)estimate mean free path and stopping powers at relativistic energies. Since the comparison calculations are performed within a Bethe theory, i.e., averaging over the momentum dependence of the Bethe ridge and only using optical data, we conclude that our momentum-dependence (Gaussian) model for the Bethe ridge is not suitable for use at relativistic energies. Therefore, we propose to use the simple Bethe theory until a more suitable model can be developed. Further investigations need to be done. We proceed in getting our tools (i.e., the Monte Carlo codes, geometry models, etc) together to address our main hypotheses. The geometry model for the bone/bone-marrow interface looks promising and scaling to mouse dimensions

	reasonable and possible. The development of interaction cross sections needed for the event-by-event Monte Carlo simulations as well as the development and extension of the codes progresses. However, the difficulties to calculate reliable cross sections at relativistic energies delay implementation of cross sections into the Monte Carlo codes and running simulations at highly relativistic energies. First simulation runs at intermediate energies have been started and the results are promising. However, we have to continue and extend the simulation runs to higher energies and to better statistics. Simulation runs are computationally very expensive and time consuming.
Bibliography Type:	Description: (Last Updated: 11/09/2018)
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